

7-8

1. If  $\frac{dy}{dx} = 4y$  and if  $y=4$  when  $x=0$ , then  $y=$

(A)  $4e^{4x}$



(B)  $e^{4x}$

(C)  $3+e^{4x}$

(D)  $4+e^{4x}$

(E)  $2x^2 + 4$

2. Bacteria in a certain culture increase at a rate proportional to the number present. If the number of bacteria doubles in three hours, in how many hours will the number of bacteria triple?

(A)  $3\ln 3/\ln 2$



(B)  $2\ln 3/\ln 2$

(C)  $\ln 3/\ln 2$

(D)  $\ln(27/2)$

(E)  $\ln(9/2)$

3.  A puppy weighs 2.0 pounds at birth and 3.5 pounds two months later. If the weight of the puppy during its first 6 months is increasing at a rate proportional to its weight, then how much will the puppy weigh when it is 3 months old?

(A) 4.2 pounds

(B) 4.6 pounds



(C) 4.8 pounds

(D) 5.6 pounds

(E) 6.5 pounds

4.  Population  $y$  grows according to the equation  $\frac{dy}{dt} = ky$ , where  $k$  is a constant and  $t$  is measured in years. If the population doubles every 10 years, then the value of  $k$  is

(A) 0.069



(B) 0.200

(C) 0.301

(D) 3.322

(E) 5.000

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5.  During a certain epidemic, the number of people that are infected at any time increases at a rate proportional to the number of people that are infected at that time. If 1,000 people are infected when the epidemic is first discovered, and 1,200 are infected 7 days later, how many people are infected 12 days after the epidemic is first discovered?

- (A) 343  
 (B) 1,343  
 (C) 1,367  
 (D) 1,400  
 (E) 2,057



6. Oil is being pumped continuously from a certain oil well at a rate proportional to the amount of oil left in the well; that is,  $dy/dt = ky$ , where  $y$  is the amount of oil left in the well at any time  $t$ . Initially there were 1,000,000 gallons of oil in the well, and 6 years later there were 500,000 gallons remaining. It will no longer be profitable to pump oil when there are fewer than 50,000 gallons remaining.
- At what rate is the amount of oil in the well decreasing when there are 600,000 gallons of oil remaining?

**Part B**

1 point is earned for answer consistent with part a.

$$\begin{aligned}\frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2\end{aligned}$$

Decreasing at  $10^5 \ln 2$  gal/year



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for answer consistent with part a.

$$\begin{aligned}\frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2\end{aligned}$$

Decreasing at  $10^5 \ln 2$  gal/year

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7. If  $\frac{dy}{dt} = ky$  and  $k$  is a nonzero constant, then  $y$  could be

(A)  $2e^{kty}$

(B)  $2e^{kt}$  ✓

(C)  $e^{kt+3}$

(D)  $kty+5$

(E)  $\frac{1}{2}ky^2 + \frac{1}{2}$

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8. Oil is being pumped continuously from a certain oil well at a rate proportional to the amount of oil left in the well; that is,  $dy/dt = ky$ , where  $y$  is the amount of oil left in the well at any time  $t$ . Initially there were 1,000,000 gallons of oil in the well, and 6 years later there were 500,000 gallons remaining. It will no longer be profitable to pump oil when there are fewer than 50,000 gallons remaining.
- (a) Write an equation for  $y$ , the amount of oil remaining in the well at any time  $t$ .
- (b) At what rate is the amount of oil in the well decreasing when there are 600,000 gallons of oil remaining?
- (c) In order not to lose money, at what time  $t$  should oil no longer be pumped from the well?

**Part A**

2 points are earned for  $y = e^{kt}$  **or** 1 point for each:  $\ln|y| = kt + C$ ,  $y = e^{kt} + C$

1 point is earned for  $C = 10^6$

1 point is earned for  $k = -\frac{\ln 2}{6}$

1 point is earned for answer

\*Max 2/5 If  $y$  is not an exponential

$$\frac{dy}{dt} = ky \quad \text{or} \quad \begin{cases} \frac{dy}{y} = k dt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$

$$t = 0 \Rightarrow C = 10^6, C_1 = \ln 10^6$$

$$\therefore y = 10^6 e^{kt}$$

$$t = 6 \Rightarrow \frac{1}{2} = e^{6k}$$

$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6} \ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$



|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|

The student response earns all of the following points:

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2 points are earned for  $y = Ce^{kt}$  **or** 1 point for each:  $\ln|y| = kt + C$ ,  $y = e^{kt} + C$

1 point is earned for  $C = 10^6$

1 point is earned for  $k = -\frac{\ln 2}{6}$

1 point is earned for answer

\*Max 2/5 If  $y$  is not an exponential

$$\frac{dy}{dt} = ky \quad \text{or} \quad \begin{cases} \frac{dy}{y} = k dt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$

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$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6} \ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$

### Part B

1 point is earned for answer consistent with part a.

$$\begin{aligned} \frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2 \end{aligned}$$

Decreasing at  $10^5 \ln 2$  gal/year



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

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1 point is earned for answer consistent with part a.

$$\begin{aligned}\frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2\end{aligned}$$

Decreasing at  $10^5 \ln 2$  gal/year

**Part C**

1 point is earned for solving  $5 \cdot 10^4$

1 point is earned for  $kt = -\ln 20$

1 point is earned for answer

$$\begin{aligned}5 \cdot 10^4 &= 10^6 e^{kt} \\ \therefore kt &= -\ln 20 \left[ = \ln \frac{1}{20} \right] \\ \therefore t &= \frac{-\ln 20}{\frac{-\ln 2}{6}} \\ &= 6 \frac{\ln 20}{\ln 2} = 6 \log_2 20 \\ &6 \frac{\ln 20}{\ln 2} \text{ years after starting}\end{aligned}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for solving  $5 \cdot 10^4$

1 point is earned for  $kt = -\ln 20$

1 point is earned for answer

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$$5 \cdot 10^4 = 10^6 e^{kt}$$

$$\therefore kt = -\ln 20 \left[ = \ln \frac{1}{20} \right]$$

$$\therefore t = \frac{-\ln 20}{\frac{-\ln 2}{6}}$$

$$= 6 \frac{\ln 20}{\ln 2} = 6 \log_2 20$$

$$6 \frac{\ln 20}{\ln 2} \text{ years after starting}$$